3-Way Passive Crossover — Design Proposal

Based on the measured woofer / midrange / tweeter files in Mätdata/ · 6 November 2026

Driver summary

Driver	Re (Ω)	Fs (Hz)	Qts	Le (mH)	Rel. dB	−3 dB range (Hz)
Woofer (bas)	7.24	37.3	0.94	0.29	-4.4	29 – 1.3k
Midrange (mid)	7.07	47.0	0.61	0.25	-4.5	55 – 9.0k
Tweeter (high)	5.12	493.1	0.34	0.09	-0.4	1.5k – >20k

T/S values are rough estimates from the impedance curves; the drivers may be box-loaded. SPL is relative / uncalibrated.

CROSSOVER AT A GLANCE

Woofer → mid: 306 Hz, LR2 (12 dB/oct)
Mid → tweeter: ~2.0 kHz, LR2
Mid polarity: reversed
Tweeter pad: 2.5 dB (1.5 Ω / 18 Ω)
Zobels: 9.1 Ω + 3.3 µF (W)
8.8 Ω + 3.3 µF (M)
6.4 Ω + 2.2 µF (T)
Predicted sum: -3.7 … +2.8 dB
300 Hz – 10 kHz
System Zmin: 5.7 Ω at 775 Hz

Why these crossover points?

- 306 Hz: the woofer and midrange have almost the same measured level around 300 Hz, and both are more than an octave from their −3 dB limits.
- ~2.0 kHz: the tweeter is flat above 1.5 kHz and 4× its Fs; the midrange's 5 kHz breakup is attenuated by the low-pass filter.
- LR2 keeps the component count low. The midrange is connected with reversed polarity, as required by LR2.
- Zobel networks make the driver loads more resistive near the crossover, so the calculated values behave closer to the simulation.

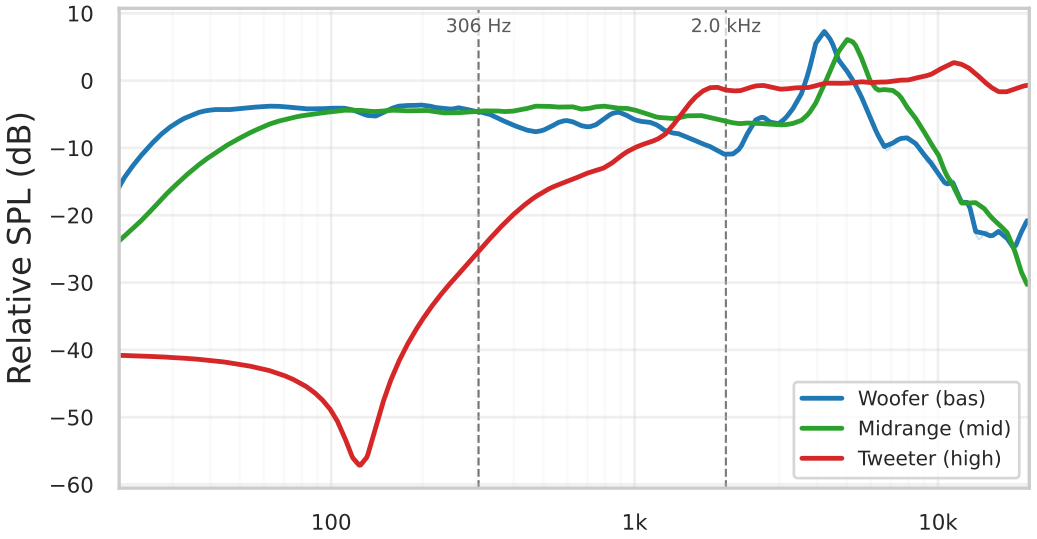
Important notes

- The SPL measurements are relative, not calibrated dB SPL. The L-pad is a starting point and may need adjustment after the final measurement.
- The impedance files may have been measured in the box. T/S values are therefore rough estimates, not free-air data.
- The simulation uses measured complex impedance and measured SPL phase. It assumes a common time reference; the final sum must be verified.
- Use low-DCR inductors, good-quality capacitors and 10 W resistors. Measure the finished system before final voicing.

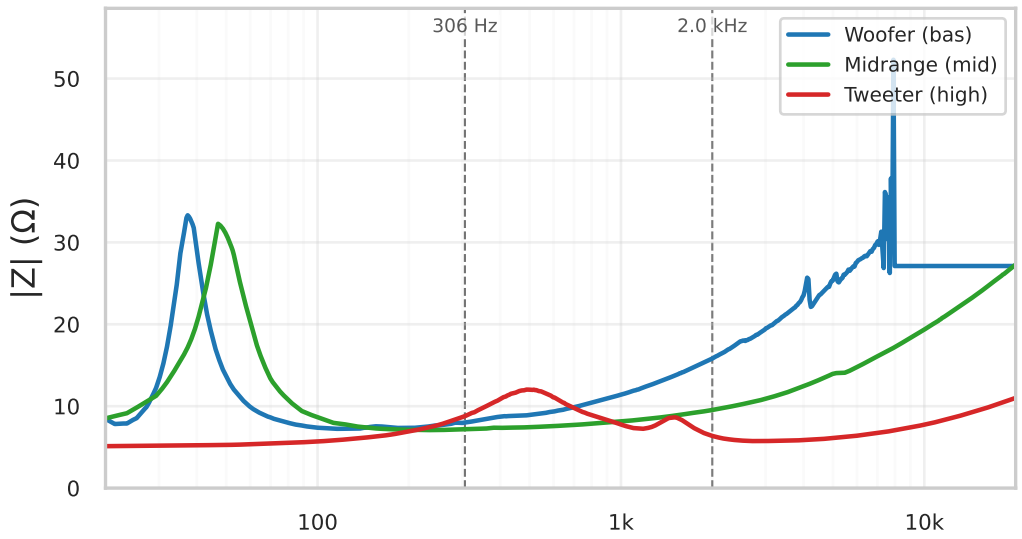
Measured driver data

SPL magnitude and phase, impedance magnitude and phase · phase unwrapped for readability

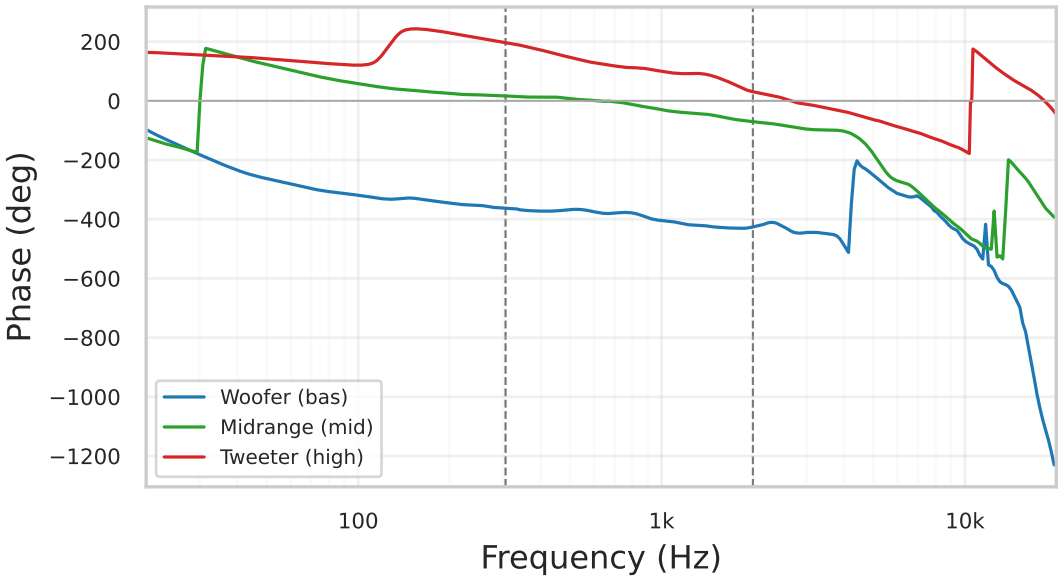
SPL magnitude (1/12-oct smoothed)



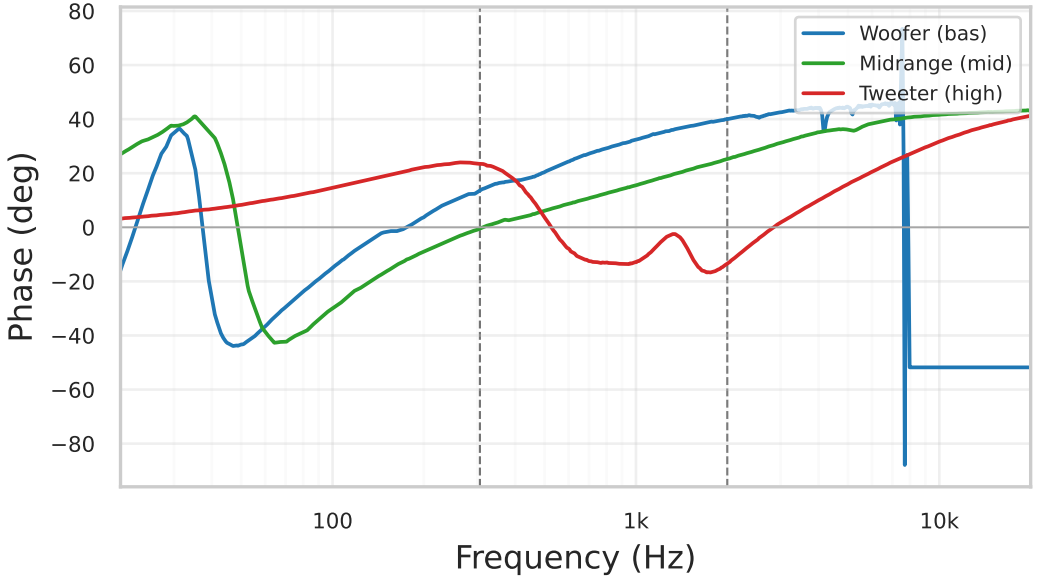
Impedance magnitude



SPL phase (unwrapped)



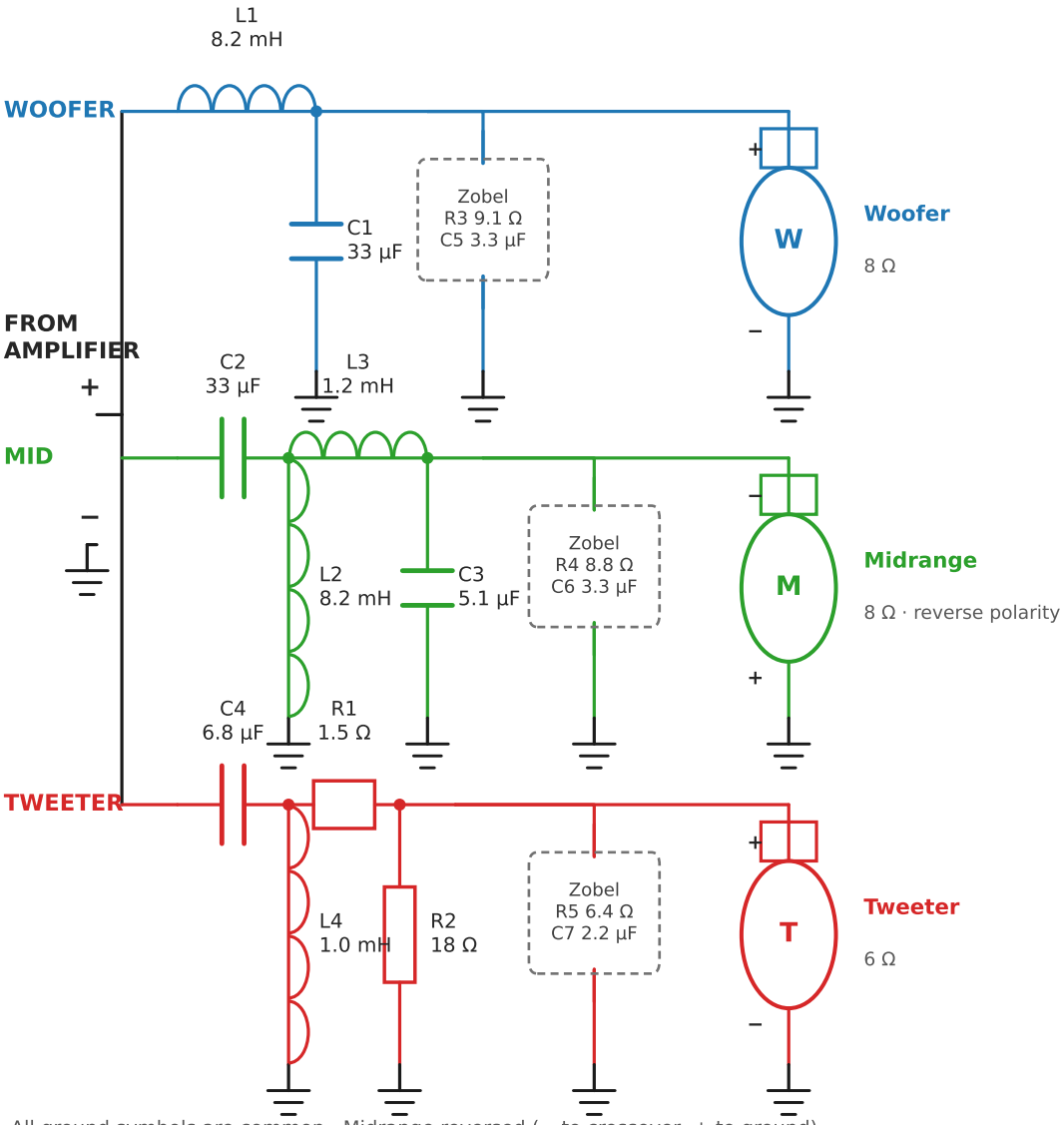
Impedance phase (unwrapped)



SPL is relative / uncalibrated. Impedance was measured with the same setup. Dashed lines mark the proposed 306 Hz and 2.0 kHz crossover points.

Proposed 3-way passive crossover

LR2 at 306 Hz and ~2.0 kHz · midrange reversed · Zobel networks included



All ground symbols are common. Midrange reversed (− to crossover, + to ground).
Zobel networks are across the driver terminals.

Component list

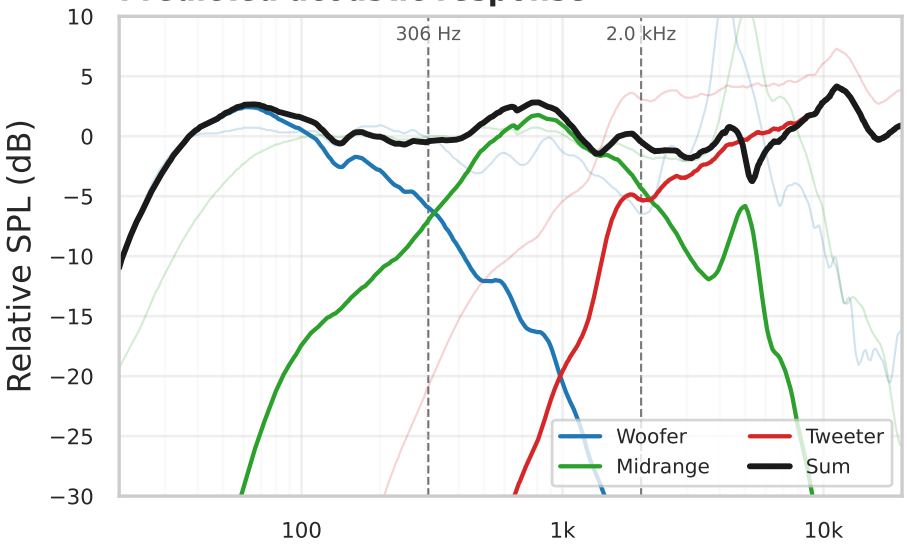
Ref	Value	Function
L1	8.2 mH	Woofer LP series
C1	33 µF	Woofer LP shunt
L2	8.2 mH	Mid HP shunt
C2	33 µF	Mid HP series
L3	1.2 mH	Mid LP series
C3	5.1 µF	Mid LP shunt
L4	1.0 mH	Tweeter HP shunt
C4	6.8 µF	Tweeter HP series
R1	1.5 Ω	Tweeter L-pad series
R2	18 Ω	Tweeter L-pad shunt
R3	9.1 Ω	Woofer Zobel
C5	3.3 µF	Woofer Zobel
R4	8.8 Ω	Mid Zobel
C6	3.3 µF	Mid Zobel
R5	6.4 Ω	Tweeter Zobel
C7	2.2 µF	Tweeter Zobel

Calculated LR2: 306 Hz (W/M), 2034 Hz (mid LP), 1930 Hz (tweeter HP). Use low-DCR inductors (L1 < 0.5 Ω; L2–L4 < 0.3 Ω), good-quality capacitors and 10 W resistors.

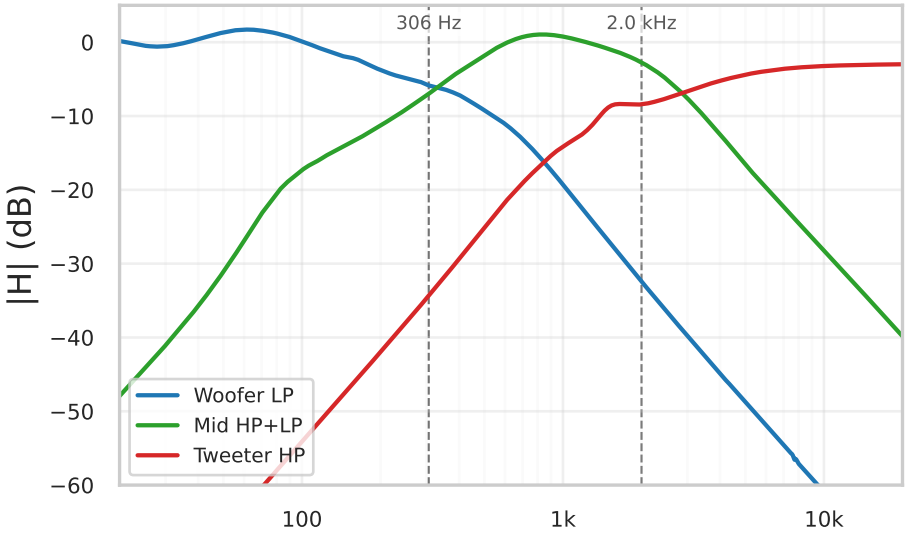
Predicted performance

Simulation using the measured complex impedance and measured SPL phase · midrange reversed

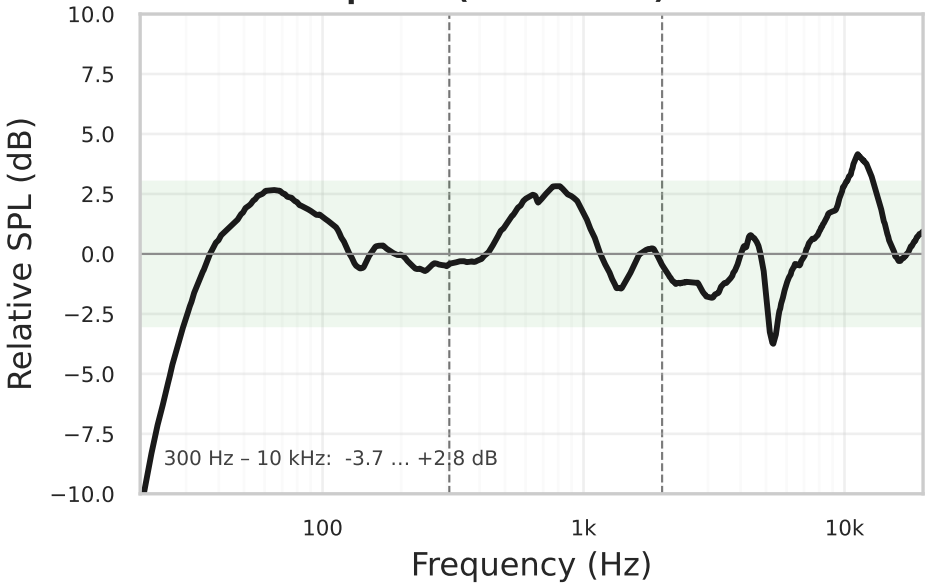
Predicted acoustic response



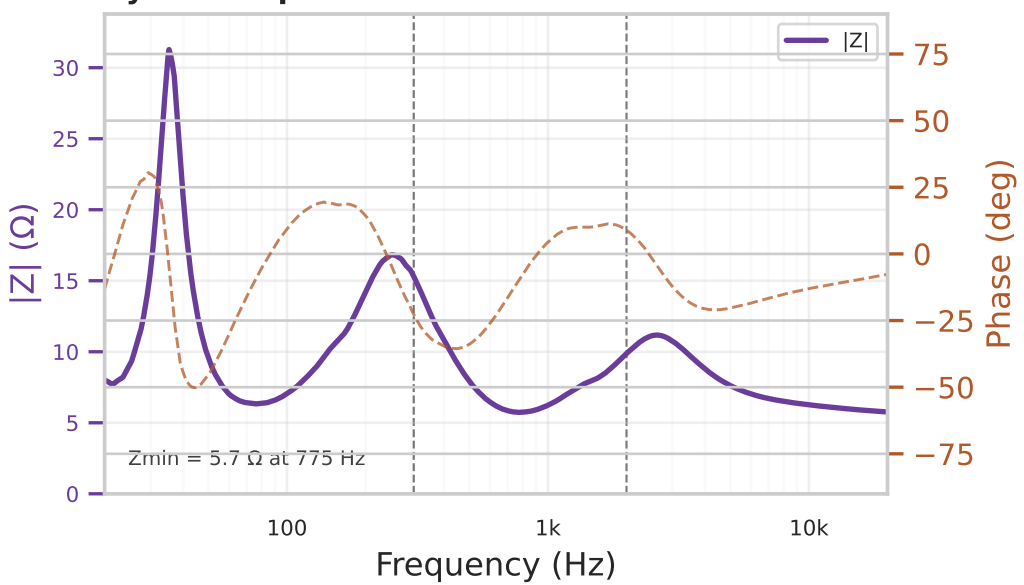
Electrical transfer functions



Summed response (normalized)



System impedance



Simulation assumes ideal drivers with the measured magnitude/phase and a common time reference. The final acoustic sum must be verified with a microphone measurement.

Bill of materials and implementation notes

Component values, wiring and a short setup checklist

Bill of materials

Ref	Value	Type / notes	Qty
L1	8.2 mH	Woofer LP series; low-DCR iron/ferrite, DCR < 0.5 Ω	1
C1	33 µF	Woofer LP shunt; MKP or high-quality bipolar electrolytic	1
L2	8.2 mH	Mid HP shunt; air-core, DCR < 0.3 Ω	1
C2	33 µF	Mid HP series; MKP or high-quality bipolar electrolytic	1
L3	1.2 mH	Mid LP series; air-core, DCR < 0.2 Ω	1
C3	5.1 µF	Mid LP shunt; MKP 100 V	1
L4	1.0 mH	Tweeter HP shunt; air-core, DCR < 0.2 Ω	1
C4	6.8 µF	Tweeter HP series; MKP 100 V	1
R1	1.5 Ω	Tweeter L-pad series; 10 W wirewound	1
R2	18 Ω	Tweeter L-pad shunt; 10 W wirewound	1
R3	9.1 Ω	Woofer Zobel; 5–10 W	1
C5	3.3 µF	Woofer Zobel; MKP	1
R4	8.8 Ω	Mid Zobel; 5–10 W	1
C6	3.3 µF	Mid Zobel; MKP	1
R5	6.4 Ω	Tweeter Zobel; 5–10 W	1
C7	2.2 µF	Tweeter Zobel; MKP	1

Wiring

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Input +  →  all three branch inputs
Input −  →  common ground

Woofer +  →  L1 output
Woofer −  →  ground

Mid −  →  crossover output
Mid +  →  ground
  (reversed polarity; or reverse the
   mid wires at the driver)

Tweeter +  →  R1 / R2 junction
Tweeter −  →  ground
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Setup checklist

1. Measure each driver's impedance in the final box.
2. Verify the crossover regions with DATS or REW.
3. Measure the summed response at 1 m on the tweeter axis.
4. Adjust R2 for treble level: 22 Ω = louder, 15 Ω = softer.
5. If the mid/tweeter sum has a dip, compare both tweeter polarities.
6. Check that the minimum impedance stays above ~4 Ω.
7. Listen before finalising; measurements are a guide.

Caveats

• SPL is relative / uncalibrated. The L-pad value is based on the relative levels in these files and may need adjustment if the absolute sensitivities differ.
• The impedance files may be box-loaded; the T/S values are rough estimates, not free-air parameters.
• Component tolerances and inductor DCR will shift the crossover frequencies. Measure the finished crossover if possible.
• This is a starting point for voicing, not a guaranteed flat system. The final acoustic result depends on the box, baffle, room and driver unit tolerances.